BCS-402	Design and Analysis of Algorithms	L T P 3 0 0	3 credits

Course Objectives:

CO1	To understand and remember algorithms and its analysis procedure.
CO2	To introduce the concept of data structures through ADT including List, Stack, Queues. Analyze the asymptotic performance of algorithms.
CO3	To design and implement various data structures and algorithms.
CO4	Apply important algorithmic design paradigms and methods of analysis.
CO5	Synthesize efficient algorithms in common engineering design situations.

MODULE-I

Introduction: Algorithms, Analyzing algorithms, Complexity of algorithms, Growth of Functions- master theorem, Sorting- insertion sort, selection sort, Shell sort, Comparison of sorting algorithms, sorting in linear time.
Divide and Conquer: General method, merge sort, quick sort, Heap sort, Strassen's matrix multiplication algorithm.
Advanced data Structures: B – trees , Red-Black trees, Data Structure for Disjoint sets.

MODULE-II

Graphs: Minimum Spanning trees – Prim's and Kruskal's algorithms, Single source shortest paths - Dijkstra's and Bellman Ford algorithms, Multistage graphs, all pairs shortest paths: Warshal's and Floyd's algorithms
The Greedy Method : optimal storage on tapes, Fractional Knapsack problem, Job sequencing with deadlines,.
Dynamic Programming: Introduction, O/1 knapsack,. Matrix chain multiplication, longest Common sequence.

MODULE-III

Back Tracking: Introduction, 8 queen's problem, graph coloring, Hamiltonian cycles, Subset Sum Problem.
Branch and Bound: Introduction, traveling salesperson problem, Euclids algorithm for GCD
String matching: Naïve String Matching, Rabin karp, Knuth-Morris-Pratt algorithm.
NP Completeness : Introduction to P, NP hard and NP completeness, NP complete problems : Clique problem, Vertex Cover problem, Travelling salesman problem.

Text Books:
1. Fundamental of Computer algorithms, Ellis Horowitz and Sartaj Sahni, 1978, Galgotia Publications
2. Introduction To Algorithms, Thomas H Cormen, Charles E Leiserson And Ronald L Rivest: , TMH

Reference Books:

1. The Design and Analysis of Computer Algorithm, Aho A.V. Hopcroft J.E., 1974, Addison Wesley.
2. Algorithms-The Construction, Proof and Analysis of Programs, Berlion, P.Bizard, P., 1986. Johan Wiley & Son
3. Introduction to Design and Analysis of Algorithm, Goodman, S.E. & Hedetnieni, 1997, MGH.

4. Introduction to Computers Science- An algorithms approach , Jean Paul Trembley, Richard B.Bunt.

Course Outcomes:

CO1	For a given algorithms analyze worst-case running times of algorithms based on asymptotic analysis and justify the correctness of algorithms .
CO2	Describe the greedy paradigm and explain when an algorithmic design situation calls for it. For a given problem develop the greedy algorithms.
CO3	Describe the divide-and-conquer paradigm and explain when an algorithmic design situation calls for it. Synthesize divide-and-conquer algorithms. Derive and solve recurrence relation.
CO4	Describe the dynamic-programming paradigm and explain when an algorithmic design situation calls for it
CO5	For a given model engineering problem, model it using graph and write the corresponding algorithm to solve the problems.